

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2019

Course: B. Tech in Chemical /Petrochemical Engineering Semester : V

Subject Name: Chemical Engineering Thermodynamics-II Subject Code: BTCHC 501

Max Marks: 60

Date: 9/12/2019

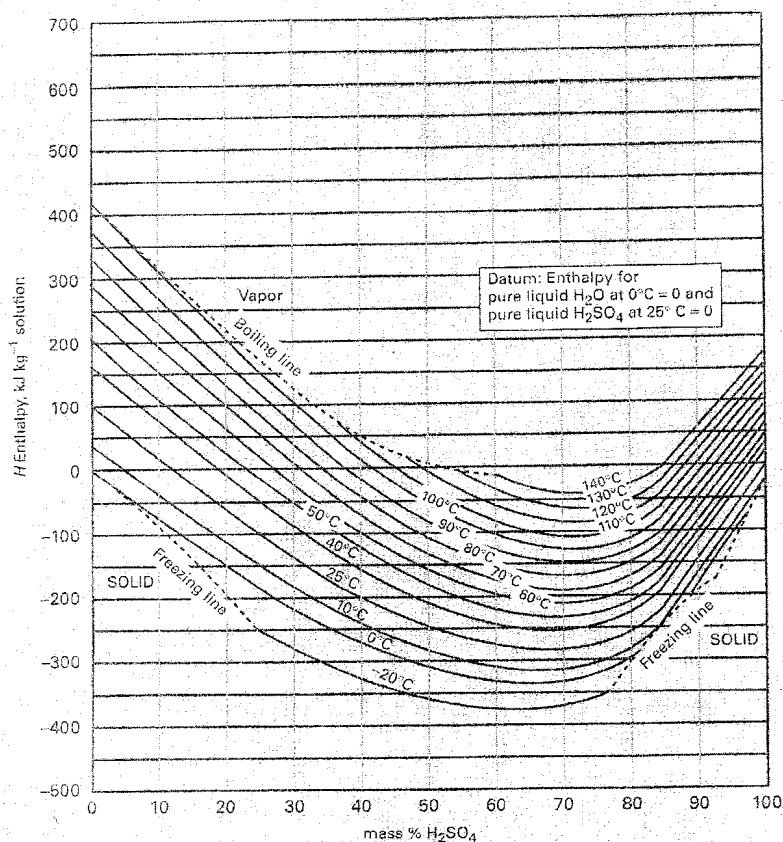
Duration: 3 Hr.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use Steam table wherever required.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) Assuming Raoult's law to be valid prepare a T-x,y diagram for a pressure of 100 kPa for the Benzene (1)/ethyl benzene(2) system. Show subcooled liquid region, superheated vapor region , and saturation temperatures of pure specie on T-x,y diagram.	(Apply)	06
B) With neat sketch , explain the concept of retrograde condensation and its application to petroleum industry.	(Understand)	06
C) A liquid mixture of cyclohexane (1)/Phenol(2) for which $x_1=0.6$ is in equilibrium with its vapor at 417.15 K. Determine the equilibrium pressure P and vapor composition y_1 from the following information.:	(Understand and apply)	06
$\ln \gamma_1 = A (x_2)^2$ $\ln \gamma_2 = A (x_1)^2$		
At 417.15K, $P_1^{sat} = 75.20$ and $P_2^{sat} = 31.66$ kPa The system forms an azeotrope at 417.15 K for which $x_1^{az} = y_1^{az} = 0.294$		
Q.2 Attempt the following.		
A) The enthalpy of binary liquid system of species 1 & 2 at T and P is given by $H = 400x_1 + 300x_1^2$ Find the value of $\bar{H}_1^\infty$ and $\bar{H}_2^\infty$.	(Apply)	06
B) Derive an expression showing that chemical potential can be used as a criteria for phase equilibrium.	(Apply)	06
Q. 3 Solve the following.		
A) The partial molar volume of methanol in a methanol (1)-water (2)solution at $x_1 = 0.3881$ is 39.176×10^{-6} m ³ /mol. The density of the mixture is 905.376 kg/m ³ . Determine the partial molar volume of water in the solution.	(Apply)	06
B) Estimate the fugacity of cyclopentane at 110 °C and 275 bar. At 110°C the vapor pressure of cyclopentane is 5.267 bar. Molar mass of cyclopentane = 70, $T_c = 511.8$ °K , $P_c = 45.02$ bar , $Z_c = 0.273$, $V_c = 258$ cm ³ /mole and $\omega = 0.196$	(Apply)	06
Q.4 A) is compulsory. Solve one from either B) or C).		
A) For the following system ,determine Margules parameters and then apply the Margules equation to sufficient number of VLE calculations to allow the construction of P-x,y diagram at a given temperature. System : 2- Butanone(1)/ Toluene (2) at 50 °C	(Apply)	08

H - x,y diagram for Q.5 A) to be attached with answer book.



Paper End

- $\gamma_1^{\text{sat}} = 1.47$ $\gamma_2^{\text{sat}} = 1.30$ $P_1^{\text{sat}} = 36.09 \text{ kPa}$ $P_2^{\text{sat}} = 12.30 \text{ kPa}$
- B) A stream of nitrogen flowing at the rate of 2 kg/s and a stream of hydrogen flowing at the rate of 0.5 kg/s mix adiabatically in a steady flow process. If the gases are assumed ideal, What is the rate of entropy increase as a result of the process? (Apply)
- C) Give the mathematical expression for Raoult's Law. State its the major assumptions. (Remember)
Give few examples of binary systems obeying Raoult's law with proper justification.
- Q. 5 Solve the following
- A) Nine thousand kg/hr of an 80 mass% H_2SO_4 solution in water at 50°C is continuously diluted with chilled water at 5°C to yield a stream containing 50 mass% H_2SO_4 at 60°C . (Apply)
i) What is the mass flow-rate of chilled water in kg/hr?
ii) What is the rate of heat transfer in kJ/hr for the mixing process? Is heat added or removed?
iii) If the mixing occurred adiabatically, what would be the temperature of the product stream? Assume here the same inlet conditions and the same product compositions as for part (ii)
- B) The equilibrium constant for the reaction $\text{A} \rightarrow \text{B}$ is doubled when the temperature is changed from 25°C to 35°C . Calculate the enthalpy change of the reaction. (Understand)
- Q.6 Solve Any Two of the following :
- A) The water gas shift reaction, (Analyze)
 $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$
is carried out under different sets of conditions described below. Calculate the fraction of steam reacted in each case. Assume the mixture behaves as an ideal gas.
i) The reactants consists of 1mol of H_2O vapor and 1 mol of CO . The temperature is 1100 K and the pressure is 1 bar.
ii) Same as i) except that pressure is 10 bar.
iii) The reactants are 2 mol of H_2O and 1 mol of CO other conditions are (assume $\ln K = 0$ for above reaction)
- B) For the ammonia synthesis reaction written : (Apply)
 $\frac{1}{2} \text{N}_2(\text{g}) + \frac{3}{2} \text{H}_2(\text{g}) \rightarrow \text{NH}_3(\text{g})$
with 0.5 mole of N_2 and 1.5 mole of H_2 as the initial amounts of reactants and with the assumption that the equilibrium mixture is an ideal gas, show that:
$$\varepsilon_e = 1 - (1 + 1.299K \frac{P}{P^0})^{-1/2}$$
- C) With neat sketch, write an informative account on fuel cell. (Remember)

Data : Antoine equation to be used is as follows :

$$\ln P^{\text{sat}} / \text{kPa} = A - \frac{B}{T / \text{K} + C}, \text{ Parameters for the Antoine Equation}$$

Chemical Specie	A	B	C
Benzene	13.8594	2773.78	-53.08
Ethyl Benzene	14.0045	3279.47	-59.95